Graphing Rational Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Reading graphs**

The graph of $f(x) = \frac{x+2}{x-3}$ is shown. From the graph and the formula, state the asymptotes and intercepts.

Vertical asymptote $x = 3$; horizontal asymptote $y = 1$; intercepts $(-2, 0)$ and $\left(0, -\frac{2}{3}\right)$.

2 **[Foundational]** • **Target: Reading graphs**

The function $g(x) = \frac{1}{x-2} + 3$ is a transformation of $y = \frac{1}{x}$.

a Describe the transformation. Shift right 2 units and up 3 units.

b State the asymptotes of g. Vertical $x = 2$; horizontal $y = 3$.

c State the domain and range. Domain $x \neq 2$; range $y \neq 3$.

(a) Shift right 2 units and up 3 units.; (b) Vertical $x = 2$; horizontal $y = 3$.; (c) Domain $x \neq 2$; range $y \neq 3$.

3 **[Foundational]** • **Target: Sign analysis**

Let $f(x) = \frac{x-1}{(x+2)(x-4)}$. Using a sign chart with critical values $x = -2, 1, 4$, determine the intervals on which $f(x) > 0$.

Testing each interval: $(-\infty, -2)$ negative; $(-2, 1)$ positive; $(1, 4)$ negative; $(4, \infty)$ positive. So $f(x) > 0$ on $(-2, 1) \cup (4, \infty)$.

4 **[Foundational]** • **Target: Sign analysis**

Solve the rational inequality $\frac{x+3}{x-2} \leq 0$.

Critical values $x = -3$ (zero) and $x = 2$ (undefined). The quotient is negative between them: $[-3, 2)$ — including -3, excluding 2.

5 **[Foundational]** • **Target: Sign analysis**

Sketch-planning: for $h(x) = \frac{2x}{x^2 - 1}$, state the asymptotes and the symmetry of the graph.

Vertical asymptotes $x = 1$ and $x = -1$; horizontal asymptote $y = 0$. Since $h(-x) = -h(x)$, the function is odd — symmetric about the origin.

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6 [Foundational] • Target: Sketching a complete graph

For $f(x) = \frac{x^2 - 1}{x^2 - 4}$, find all intercepts and asymptotes, then describe the graph's behaviour near each vertical asymptote.

x-intercepts: $x = \pm 1$. y-intercept: $f(0) = \frac{-1}{-4} = \frac{1}{4}$. Vertical asymptotes $x = \pm 2$; horizontal asymptote $y = 1$ (equal degrees). Near $x = 2$: $f \to +\infty$ as $x \to 2^+$ and $f \to -\infty$ as $x \to 2^-$. Near $x = -2$: by symmetry (the function is even), $f \to -\infty$ as $x \to -2^+$ and $f \to +\infty$ as $x \to -2^-$.

7 [Foundational] • Target: Sketching a complete graph

Sketch-planning: the function $g(x) = \frac{3}{x-1} - 2$ has a vertical asymptote at $x = 1$ and horizontal asymptote $y = -2$. Find its x- and y-intercepts.

y-intercept: $g(0) = \frac{3}{-1} - 2 = -5$. x-intercept: $0 = \frac{3}{x-1} - 2$ gives $\frac{3}{x-1} = 2$, so $x - 1 = \frac{3}{2}$, $x = \frac{5}{2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Building a rational function from a description

Construct a rational function with vertical asymptote $x = 2$, horizontal asymptote $y = 1$, and a hole at $x = -3$.

Start with a base of $\frac{x}{x-2}$ (horizontal asymptote $y = 1$, vertical asymptote $x = 2$), then introduce a common factor $(x + 3)$ in both numerator and denominator: $f(x) = \frac{x(x+3)}{(x-2)(x+3)}$.

9 [Intermediate] • Target: Graphing with a slant asymptote

For $f(x) = \frac{x^2 - 4}{x - 1}$, find the slant asymptote and the vertical asymptote, and use them together with intercepts to describe the overall shape of the graph.

Dividing: $x^2 - 4 = (x - 1)(x + 1) - 3$, so $f(x) = x + 1 - \frac{3}{x-1}$. Slant asymptote: $y = x + 1$. Vertical asymptote: $x = 1$. x-intercepts: $x = \pm 2$. As $x \to \pm\infty$, the graph hugs the line $y = x + 1$, while near $x = 1$ it shoots off to $\pm\infty$.

10 [Intermediate] • Target: Full graph analysis: putting it all together

For $f(x) = \frac{2x^2 - 2}{x^2 - 9}$, find all intercepts, all asymptotes, and describe the symmetry of the graph (even, odd, or neither).

x-intercepts: $2x^2 - 2 = 0 \Rightarrow x = \pm 1$. y-intercept: $f(0) = \frac{-2}{-9} = \frac{2}{9}$. Vertical asymptotes: $x = \pm 3$. Horizontal asymptote (equal degrees): $y = 2$. Symmetry: $f(-x) = \frac{2x^2 - 2}{x^2 - 9} = f(x)$, so the function is EVEN, symmetric about the y-axis.

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11 **[Intermediate]** • **Target: Graphing near a hole**

For $f(x) = \frac{x^2 - 9}{x - 3}$, describe how the graph looks immediately around $x = 3$, and find the coordinates of the hole.

Simplify: $f(x) = x + 3$ for $x \neq 3$. The graph is just the line $y = x + 3$ with a single point removed at $x = 3$: the hole is at $(3, 6)$ — an open circle on an otherwise straight line.

12 **[Intermediate]** • **Target: Graphing near a hole**

A rectangular garden's cost per square meter to fence is modeled by $C(x) = \frac{200}{x} + 8x$, where x is the garden's width in meters. Find the vertical asymptote and horizontal/slant asymptote of this function, and interpret what the vertical asymptote means physically (why $x = 0$ isn't a realistic input).

Vertical asymptote: $x = 0$. As $x \to 0^+$, $C(x) \to \infty$ — a garden of near-zero width would require infinite fencing cost per unit length, which isn't physically meaningful; this confirms $x = 0$ must be excluded from the domain. There is no horizontal asymptote (the $8x$ term grows without bound), and the function behaves like $y = 8x$ for large x (a slant-like trend, though not a strict polynomial slant asymptote since $\frac{200}{x} \to 0$).

13 **[Intermediate]** • **Target: Graphing near a hole**

State the domain of $f(x) = \frac{\sqrt{x}}{x - 4}$.

Need $x \geq 0$ (for the square root) and $x \neq 4$ (denominator). Domain: $[0, 4) \cup (4, \infty)$.

14 **[Intermediate]** • **Target: Building a rational function from a description**

Level 2 · MediumThe function $f(x) = x^3 - 3x^2 - 9x + 2$ has a local maximum and a local minimum. Determine the intervals on which f is concave up and concave down, and find the coordinates of the inflection point.

The rate of change switches from decreasing to increasing at the inflection point. Taking the second derivative (or analyzing symmetric rates of change): $f'(x) = 3x^2 - 6x - 9, \quad f''(x) = 6x - 6 = 0 \Rightarrow x = 1.$ For x . For $x > 1$, $f''(x) > 0$, so f is concave up on $(1, \infty)$. The inflection point is $(1, f(1)) = (1, 1 - 3 - 9 + 2) = \mathbf{(1, -9)}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging] • Target: Graphing with a slant asymptote • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations**

The function $f(x) = \frac{4x-7}{2x+8}$ has horizontal asymptote $y = H$ and vertical asymptote $x = V$. Determine the value of $H + V$.

Step 1: Find horizontal asymptote $y = H$: Since degrees of numerator and denominator are both 1, the horizontal asymptote is the ratio of leading coefficients: $H = \frac{4}{2} = 2$. Step 2: Find vertical asymptote $x = V$: Set denominator to zero: $2x + 8 = 0 \Rightarrow 2x = -8 \Rightarrow V = -4$. Step 3: Compute sum: $H + V = 2 + (-4) = \mathbf{-2}$.

16 **[Challenging] • Target: Full graph analysis: putting it all together • Cross-Lesson: 4.01 Circles and Parabolas**

Solve the rational inequality $\frac{x+4}{x-2} \leq 3$ and express the solution set in interval notation.

Step 1: Subtract 3 and combine over a single common denominator: $\frac{x+4}{x-2} - 3 \leq 0 \Rightarrow \frac{(x+4)-3(x-2)}{x-2} \leq 0 \Rightarrow \frac{-2x+10}{x-2} \leq 0$. Step 2: Factor and identify critical boundary values: $\frac{-2(x-5)}{x-2} \leq 0 \Leftrightarrow \frac{x-5}{x-2} \geq 0$. Critical values are $x = 2$ (undefined) and $x = 5$ (zero). Step 3: Sign chart analysis: The quotient $\frac{x-5}{x-2}$ is positive on $(-\infty, 2)$ and $(5, \infty)$, and negative on.

17 **[Challenging] • Target: A compound rational inequality • Cross-Lesson: 4.02 Ellipses**

Level 3 · AdvancedExperimental laboratory data pairs (x, y) are transformed by computing $Y = \ln(y)$. A linear regression performed on (x, Y) yields the best-fit line $Y = -0.42x + 2.85$. Write the corresponding non-linear model for y as an exponential function of x in the form $y = ab^x$, giving a and b to three decimal places. Then predict the value of y when $x = 5$.

Step 1 (Exponentiate the Linear Equation): $\ln(y) = -0.42x + 2.85 \Rightarrow y = e^{-0.42x+2.85} = e^{2.85} \cdot e^{-0.42x} = e^{2.85} \cdot (e^{-0.42})^x$. Step 2 (Compute Parameters): $a = e^{2.85} \approx 17.28775 \approx 17.288$, $b = e^{-0.42} \approx 0.65705 \approx 0.657$. The exponential model is $y = 17.288(0.657)^x$ (or $y = 17.288e^{-0.42x}$). Step 3 (Predict for x = 5):

$y(5) = 17.288 \cdot e^{-0.42 \times 5} = 17.288 \cdot e^{-2.1} \approx 17.288 \cdot 0.122456 \approx \mathbf{2.117}$.

18 **[Challenging] • Target: Graphing near a hole • Cross-Lesson: 4.03 Hyperbolas**

Find all real and complex roots of the polynomial $P(x) = 2x^3 + 3x^2 - 8x + 3$.

By the Rational Root Theorem, possible rational roots are $\pm\{1, 3\}/\{1, 2\} = \pm\{1, 3, 1/2, 3/2\}$. Test $x = 1$: $P(1) = 2(1)^3 + 3(1)^2 - 8(1) + 3 = 2 + 3 - 8 + 3 = 0$. ✓ So $(x - 1)$ is a factor. Divide $P(x)$ by $(x - 1)$ using synthetic division: $2x^3 + 3x^2 - 8x + 3 = (x - 1)(2x^2 + 5x - 3)$. Factor the remaining quadratic: $2x^2 + 5x - 3 = (2x - 1)(x + 3)$. Setting each factor to zero yields the complete set of roots: $x = 1, \quad x = \frac{1}{2}, \quad x = -3$.

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19 [Challenging] • Target: Reading graphs • Cross-Lesson: 5.01 Vectors: Components and Operations

Solve the radical equation $\sqrt{2x+7} - x = 2$ for all real solutions and confirm extraneous roots.

Isolate the radical term: $\sqrt{2x+7} = x+2$. Square both sides: $2x+7 = (x+2)^2 = x^2+4x+4$. $x^2+2x-3 = 0 \Rightarrow (x+3)(x-1) = 0 \Rightarrow x = -3$ or $x = 1$. Test in the original un-squared equation:Check $x = -3$: $\sqrt{2(-3)+7} - (-3) = \sqrt{1}+3 = 1+3 = 4eq2$. (Extraneous ✗) Check $x = 1$: $\sqrt{2(1)+7} - 1 = \sqrt{9} - 1 = 3 - 1 = 2$. (Valid ✓) The only valid real solution is $x = 1$.

20 [Challenging] • Target: Sign analysis • Cross-Lesson: 5.02 The Dot Product

Find the set of values of x satisfying the inequality $2x^2 - 7x + 3 \leq 0$.

Find the critical values where $2x^2 - 7x + 3 = 0$. Factor the quadratic: $(2x-1)(x-3) = 0$, so the roots are $x = \frac{1}{2}$ and $x = 3$. Since the coefficient of x^2 is positive ($a = 2 > 0$), the parabola opens upward. The quadratic is less than or equal to zero between and including the roots. Therefore, the solution set is $\frac{1}{2} \leq x \leq 3$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Reading graphs

Which function has a slant asymptote?

a. $f(x) = \frac{1}{x^2}$

b. $f(x) = \frac{x}{x^2+1}$

c. $f(x) = \frac{x^2+1}{x}$ ✓

d. $f(x) = \frac{x+1}{x+2}$

Degree numerator = degree denominator + 1: slant asymptote.

22 [Foundational] • Target: Sign analysis

Which function has no vertical asymptote?

a. $f(x) = \frac{x}{x-1}$

b. $f(x) = \frac{1}{x^2}$

c. $f(x) = \frac{x^2+1}{x^2+1} = 1$ ✓

d. $f(x) = \frac{1}{x}$

No denominator zero possible when simplified.

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23 [Foundational] • Target: Sketching a complete graph

Find all rational functions with vertical asymptotes at $x = \pm 1$ and horizontal asymptote $y = 0$.

a. $\frac{x}{x^2 - 1}$ only

b. $\frac{1}{x - 1}$ only

c. Any polynomial

d. **Functions of the form $\frac{k}{x^2 - 1}$ ✓**

$\frac{k}{(x - 1)(x + 1)}$ for any nonzero k.

24 [Foundational] • Target: More rational inequalities

Which is the graph of an even function?

a. **Symmetric about the y-axis ✓**

b. Symmetric about the origin

c. Symmetric about the x-axis

d. Has no symmetry

Even: $f(-x) = f(x)$ — y-axis symmetry.

25 [Foundational] • Target: Building a rational function from a description

What is the maximum value of the function $f(x) = \sqrt{3}\sin(2x) - \cos(2x) - 2\sin x + 2\sqrt{3}\cos x$ for all real numbers x?

a. **3 ✓**

b. 2

c. 6

d. 1

We solve the problem by synthesizing harmonic form (linear combination of sine and cosine), angle-doubling relationships, and quadratic optimization on a restricted domain. **Step 1: Express the single-angle terms in harmonic form.** Recall the linear combination identity $A\cos x + B\sin x = R\cos(x - \alpha)$. For the terms $2\sqrt{3}\cos x - 2\sin x$:
$2\sqrt{3}\cos x - 2\sin x = 4\left(\frac{\sqrt{3}}{2}\cos x - \frac{1}{2}\sin x\right)$ Using the cosine addition formula.

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26 [Foundational] • Target: Graphing with a slant asymptote

A function f is defined for all real numbers x such that $x \neq 0$ and $x \neq 1$. If $f(x) + 2f\left(\frac{1}{1-x}\right) = 3x$ for all permissible values of x, what is the value of $f(3)$?

a. $\frac{20}{9}$ ✓

b. $\frac{7}{3}$

c. $\frac{16}{9}$

d. 3

To find $f(3)$, we use the cyclic substitution property of the transformation $T(x) = \frac{1}{1-x}$. 1. Substitute $x = 3$ into the given functional equation: $f(3) + 2f\left(\frac{1}{1-3}\right) = 3(3) \Rightarrow f(3) + 2f\left(-\frac{1}{2}\right) = 9 \quad - - -$ (Equation 1) 2. Next, evaluate the equation at $x = -\frac{1}{2}$: $\frac{1}{1-(-1/2)} = \frac{1}{3/2} = \frac{2}{3}$.

27 [Foundational] • Target: Full graph analysis: putting it all together

The function f is defined for all real numbers $x \neq -2$ by $f(x) = \frac{4x-1}{x+2}$ What is the value of $f^{-1}(3)$?

a. -5

b. 3

c. 5

d. 7 ✓

By definition of the inverse function, if $f^{-1}(3) = x$, then $f(x) = 3$. Set the function equal to 3 and solve for x: $\frac{4x-1}{x+2} = 3$ Multiply both sides by $x + 2$: $4x - 1 = 3(x + 2)$ $4x - 1 = 3x + 6$ Subtract $3x$ from both sides and add 1: $4x - 3x = 6 + 1$ $x = 7$ Thus, $f^{-1}(3) = 7$. Common Misconceptions: - Choice A (-5) arises from a sign error when clearing the denominator as $3(x - 2)$, yielding $4x - 1 = 3x - 6 \Rightarrow x = -5$. - Choice B (3) results from forgetting to dist.

28 [Foundational] • Target: A compound rational inequality

Which function is decreasing on all of $\mathbb{R}$?

a. $y = e^x$

b. $y = x^2$

c. $y = e^{-x}$ ✓

d. $y = |x|$

Applying Unit 4 · Functions Involving Parameters, Vectors, and Matrices principles, e^{-x} decreases everywhere. Therefore, the verified correct option is $y = e^{-x}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Graphing near a hole

Find all solutions of $\frac{x^2+2x}{x^2-1} = 1 + \frac{3}{x-1}$.

a. $x = 1$

b. No solution

c. $x = -4$

d. $x = 4$ ✓

LCD $= (x-1)(x+1)$: $x^2+2x = (x^2-1)+3(x+1) \Rightarrow x^2+2x = x^2+3x+2 \Rightarrow -x = 2 \Rightarrow x = -2$. Check: denom nonzero at $x = -2$. Answer: $x = -2$.

30 [Intermediate] • Target: Reading graphs

Vertical asymptote of $\frac{x+1}{x-4}$:

a. $x = 1$

b. $x = 4$ ✓

c. $x = -1$

d. $x = -4$

Applying Unit 1 · Polynomial and Rational Functions principles, Denominator = 0 at $x = 4$. Therefore, the verified correct option is $x = 4$.

31 [Intermediate] • Target: Sign analysis

Identify: $\frac{p(x)}{q(x)}$ is called a _____ function.

a. Irrational

b. **Rational** ✓

c. Polynomial

d. Transcendental

Applying Rational Expressions & Functions principles, By definition. Therefore, the verified correct option is Rational.

32 [Intermediate] • Target: Sketching a complete graph

Describe the graph of $f(x) = x^3(x-2)^2$ near $x = 0$ and $x = 2$.

a. Crosses at both

b. **Crosses at $x = 0$; touches at $x = 2$** ✓

c. Touches at $x = 0$; crosses at $x = 2$

d. Touches at both

Multiplicity 3 (odd): crosses at $x = 0$. Multiplicity 2 (even): touches at $x = 2$.

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33 [Intermediate] • Target: More rational inequalities

A function is defined by $f(x) = 2x - 4$ for all x near $x = -4$ (including at $x = -4$). What is $\lim_{x \to -4} f(x)$?

a. -15

b. -12 ✓

c. -14

d. -10

f is a single continuous linear rule here, so the limit equals $f(-4) = -12$.

34 [Intermediate] • Target: Building a rational function from a description

A rational function $\frac{P(x)}{Q(x)}$ where $\deg P > \deg Q$ may have:

a. No asymptote

b. A horizontal asymptote $y = 1$

c. **An oblique (slant) asymptote** ✓

d. A horizontal asymptote $y = 0$

Degree of numerator exactly 1 more than denominator: slant asymptote.

35 [Intermediate] • Target: Graphing with a slant asymptote

Find all rational zeros of $P(x) = x^3 - 7x + 6$.

a. $x = 1, 2, -3$ ✓

b. $x = 1, -2, 3$

c. $x = -1, -2, 3$

d. $x = -1, 2, 3$

Possible: $\pm 1, \pm 2, \pm 3, \pm 6$. Test: $P(1) = 0, P(2) = 0, P(-3) = 0$.

36 [Intermediate] • Target: Full graph analysis: putting it all together

Find the oblique asymptote of $f(x) = \frac{x^2 + 1}{x}$.

a. $y = 0$

b. $y = x + 1$

c. $y = x$ ✓

d. $y = 1$

$f(x) = x + \frac{1}{x}$. As $x \to \infty$, $y \approx x$.

37 [Intermediate] • Target: A compound rational inequality

The polynomial $p(x) = x^4 - 5x^2 + 4$. Which of the following are all its real zeros?

a. $x = \pm 1, \pm 2$ ✓

b. $x = \pm 2$

c. $x = 1, 2$

d. $x = \pm 1$

Factor: $(x^2-1)(x^2-4) = (x-1)(x+1)(x-2)(x+2)$. Zeros: $\pm 1, \pm 2$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Graphing near a hole • Cross-Lesson: 4.03 Hyperbolas

Find all x-intercepts of $f(x) = \frac{x^2 - 4}{x + 3}$.

a. $x = \pm 2$ ✓

b. $x = -3$

c. $x = 4$

d. $x = 2$

Set numerator to zero: $x = \pm 2$ (denominator $\neq 0$ there).

39 [Challenging] • Target: Reading graphs • Cross-Lesson: 5.01 Vectors: Components and Operations

If $f(x) = \frac{P(x)}{Q(x)}$ with $\deg P = \deg Q$, what is the horizontal asymptote?

a. No horizontal asymptote

b. $y = \frac{\text{leading coeff of } P}{\text{leading coeff of } Q}$ ✓

c. $y = 0$

d. $y = 1$

Equal degrees: ratio of leading coefficients.

40 [Challenging] • Target: Sign analysis • Cross-Lesson: 5.02 The Dot Product

Find all values excluded from the domain of $f(x) = \frac{x^2 - 1}{x^3 - x}$.

a. $x = 0, \pm 1$ ✓

b. $x = 0$

c. $x = \pm 1$

d. $x = 1$

$x^3 - x = x(x - 1)(x + 1) = 0 \Rightarrow x = 0, \pm 1$.

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41 [Challenging] • Target: Sketching a complete graph • Cross-Lesson: 5.03 Parametric Equations

A function f is measurable and satisfies $f(x+y)=f(x)+f(y)$ a.e. Then:

a. f may be discontinuous

b. $f(x)=cx$ **a.e.** ✓

c. $f(x)=0$ a.e.

d. $f(x)=cx^2$

Measurable additive functions are linear.

42 [Challenging] • Target: More rational inequalities • Cross-Lesson: 6.01 Polar Coordinates and Conversions

The graph of $y=\frac{1}{x-2}$ is the graph of $y=\frac{1}{x}$ shifted:

a. **Right 2 units** ✓

b. Down 2 units

c. Up 2 units

d. Left 2 units

Replacing x with $x-2$ shifts right.

43 [Challenging] • Target: Building a rational function from a description • Cross-Lesson: 6.02 Polar Equations and Graphs

Determine all values of x excluded from the domain of $\frac{x^3+1}{x^2+x}$.

a. $x=-1$ only

b. All real x

c. $x=0,\ -1$ ✓

d. $x=0,1$

$x^2+x=x(x+1)=0\Rightarrow x=0,\ -1$.

44 [Challenging] • Target: Graphing with a slant asymptote • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

A quadratic function is given by $f(x)=x^2-4x+3$. The graph of this function is a parabola. Which of the following points lies on the graph of $f(x)$?

a. $(1,0)$ ✓

b. $(0,1)$

c. $(2,3)$

d. $(4,4)$

To determine which point lies on the graph of $f(x)=x^2-4x+3$, we substitute the x-coordinate of each point into the function and check if the resulting y-coordinate matches the y-coordinate of the point. For the point $(1,0)$: Substitute $x=1$ into $f(x)$: $f(1)=(1)^2-4(1)+3=1-4+3=0$. Since $f(1)=0$, the point $(1,0)$ lies on the graph. For the point $(0,1)$: Substitute $x=0$ into $f(x)$: $f(0)=(0)^2-4(0)+3=0-0+3=3$. Since $f(0)=3eq1$, the point $(0,1)$ does not lie.

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45 [Challenging] • Target: Full graph analysis: putting it all together • Cross-Lesson: 7.02 Logarithms and Their Properties

The rational function f is defined by $f(x) = \frac{2x^4 - 3x^3 + kx^2 - 8x + 3}{(x-2)(x^2+1)}$, where k is a real constant. Given that the series expansion of $f(x)$ in ascending powers of x, valid for sufficiently small $|x|$, has no term in x^2, find the value of k.

a. $\frac{25}{4}$ ✓

b. $\frac{17}{4}$

c. $-\frac{25}{4}$

d. $\frac{115}{12}$

To find the series expansion of $f(x)$, we first decompose $f(x)$ into partial fractions. **Step 1: Algebraic Division** The degree of the numerator is 4 and the degree of the denominator is $1 + 2 = 3$. Because the numerator has a higher degree than the denominator, $f(x)$ is an improper algebraic fraction and polynomial division must be carried out first. Expanding the denominator:
$(x-2)(x^2+1) = x^3 - 2x^2 + x - 2$. Dividing $2x^4 - 3x^3 + kx^2 - 8x + 3$ by $x^3 - 2x^2 + x - 2$.

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